

Restaurant Revenue Prediction using Apache Spark and MongoDB

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Problem Statement

- Predict the future revenue of restaurants based on various features (e.g., Rating, Budget, Cuisine, Ambience, Location).
- Build and compare machine learning models (Linear Regression, Random Forest, GBT) to find the most accurate prediction method.
- Use big data tools like Apache Spark and MongoDB to handle data ingestion, processing, and model training at scale.

Prediction Type Justification

- Understanding the Prediction Task
- Why We Use Regression (Not Classification/Clustering)
- Model Type Comparison: Prediction vs Others

Task Type	Purpose	Output Example	Used in this Project?
Prediction (Regression)	Predict a continuous value	\$ 432,000 (Revenue)	Yes
Classification	Predict a category/label	"High Revenue", "Low"	No
Clustering	Group similar items (no labels)	Cluster 1, Cluster 2	No

Technologies Used

- **Apache Spark 3.4.1 (Java API)**
Used for distributed data processing and machine learning (MLlib)
- **Apache Spark MLlib**
Built-in machine learning library used to train models like Linear Regression, Random Forest, and Gradient Boosted Trees
- **MongoDB**
Used as the data source via Spark MongoDB Connector ([mongo-spark-connector](#))
- **Java 11**
Language used for implementation
- **Apache Maven**
Used for dependency and build management ([pom.xml](#))
- **IntelliJ IDEA**
Development IDE used to write and run Spark Java code

Maven

What is Maven?

- A build automation and dependency management tool for Java projects.
- Uses a `pom.xml` file to manage project structure, versions, and libraries.

Why used here?

- Simplifies adding Spark and MongoDB dependencies.
- Automatically downloads required libraries (e.g., Spark Core, MLlib).
- Ensures consistent builds across systems and developers.
- Works seamlessly with IntelliJ IDEA.

Spark MLlib

What is MLlib?

- Spark's built-in **machine learning library**.

Why used in this project?

- Enables training of models like:
 - Linear Regression
 - Random Forest Regressor
 - Gradient Boosted Trees (GBT)

IntelliJ IDEA

What is IntelliJ?

- A powerful Java IDE used for writing and running Spark code.

Why used in this project?

- Excellent Maven integration (auto-resolves `pom.xml`)
- Smart code suggestions and debugging tools
- Easy to run local Spark applications
- Helps manage large codebases efficiently

RDD

What is RDD?

- **Resilient Distributed Dataset** — Spark's core data structure.

Key Features:

- Immutable, distributed collection of objects
- Supports fault tolerance and parallel processing
- Useful for low-level transformations (e.g., `map`, `filter`)

Note:

- In this project, we used **DataFrames** (built on RDDs) for ML tasks.

Dataset and Source

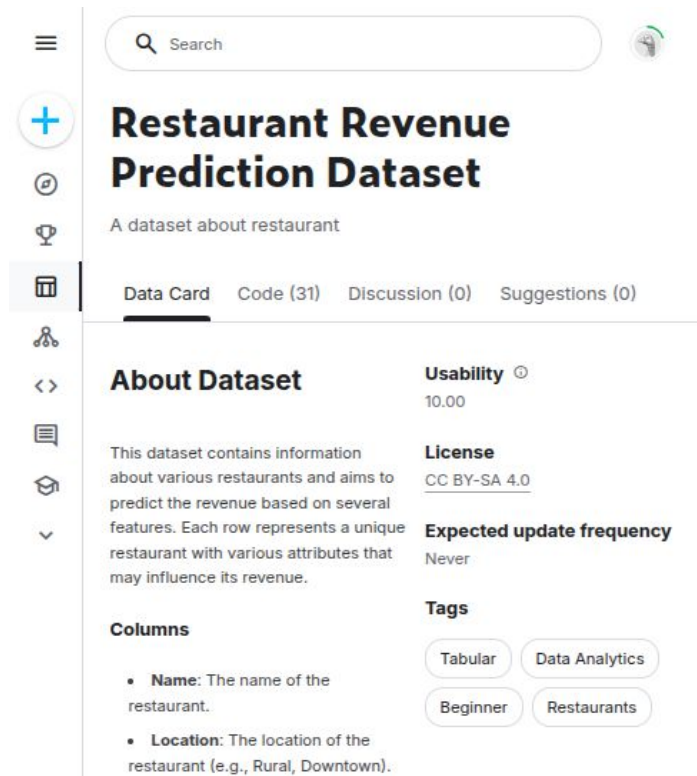
Dataset source:

 [Kaggle - Restaurant Revenue Prediction Dataset](#)

Original CSV was imported into **MongoDB**, then read using Spark

Dataset contains **restaurant-level features**, such as:

- Ratings, Marketing Budget, Cuisine, Location
- Ambience, Chef Experience, Social Media, Reviews



The screenshot shows the Kaggle dataset page for 'Restaurant Revenue Prediction Dataset'. The page has a sidebar on the left with navigation icons. The main content area includes a search bar, the dataset title, a description, and tabs for 'Data Card', 'Code (31)', 'Discussion (0)', and 'Suggestions (0)'. The 'Data Card' tab is active, showing 'About Dataset' information, 'Usability' (10.00), 'License' (CC BY-SA 4.0), 'Expected update frequency' (Never), 'Tags' (Tabular, Data Analytics, Beginner, Restaurants), and 'Columns' (Name, Location).

Search

Restaurant Revenue Prediction Dataset

A dataset about restaurant

Data Card Code (31) Discussion (0) Suggestions (0)

About Dataset

This dataset contains information about various restaurants and aims to predict the revenue based on several features. Each row represents a unique restaurant with various attributes that may influence its revenue.

Usability ⓘ
10.00

License
[CC BY-SA 4.0](#)

Expected update frequency
Never

Tags

Tabular Data Analytics Beginner Restaurants

Columns

- **Name:** The name of the restaurant.
- **Location:** The location of the restaurant (e.g., Rural, Downtown).

Features Used

Numerical Features:

- Rating
- Marketing Budget
- Number of Reviews
- Average Meal Price
- Ambience Score
- Chef Experience Years
- Service Quality Score
- Social Media Followers
- **Interaction Features:**
 - Rating × Budget
 - Ambience × Chef

Categorical Features:

- Cuisine
- Location
- Parking Availability

Data Preprocessing

- Removed missing/null values
- Filtered out extreme outliers (Revenue > \$5,00,000)
- Encoded categorical columns using `StringIndexer` & `OneHotEncoder`
- Created interaction features:
 - `Rating × Budget`
 - `Ambience × Chef`
- Applied log transformation on Revenue for GBT model

Model Training

Compared 3 models using Spark MLlib:

- Linear Regression
- Random Forest Regressor
- Gradient Boosted Trees (GBT)

Used `VectorAssembler`, `Pipeline`, and evaluation metrics

GBT model included feature engineering and log-transformed labels

```
GBTRegressor gbt = new GBTRegressor()
    .setLabelCol("RevenueLog")
    .setFeaturesCol("features")
    .setMaxDepth(3)
    .setMaxIter(40);

Pipeline pipeline = new Pipeline().setStages(new PipelineStage[]{
    indexers[0], indexers[1], indexers[2],
    encoders[0], encoders[1], encoders[2],
    assemblerGBT, gbt
});
```

Linear Regression

What is Linear Regression?

- A model that predicts a continuous output by fitting a line : $y=mx+c$

Why used here?

- Simple and fast baseline model
- Helps understand the relationship between features (e.g., budget, rating) and revenue

Limitation:

- May underperform if relationships are non-linear

Gradient Boosted Trees

What is GBT?

- An **ensemble model** of decision trees trained sequentially to reduce error.

Why used here?

- Handles complex, non-linear patterns well
- Most accurate in our case (best RMSE & R^2)

Enhancements Used:

- Log-transform on revenue
- Feature interactions (e.g., **Rating** × **Budget**)

Models Compared:

Model	Description	Strengths
Linear Regression	Predicts revenue based on weighted input features	Simple, fast, interpretable
Random Forest	Ensemble of decision trees	Handles non-linear patterns well
GBT (Gradient Boosted Trees)	Boosted tree-based model (log-transformed revenue)	Most accurate, good generalization

Prediction Output

- Predictions were made using the trained GBT model
- The revenue was **log-transformed during training** and reverted back before comparison
- We compare the model's output to the actual revenue for validation

```
+-----+-----+
| Revenue| prediction_actual|
+-----+-----+
|372361.37|414586.29792630515|
|261326.89| 323475.3443045785|
|341552.37| 313807.3024190266|
|452932.93| 376970.3216156551|
|396465.45| 382376.3099568078|
|405786.88| 410985.0193034364|
|300684.68| 359206.336752205|
|398573.67|433282.37190142105|
|385802.15|411961.59999692027|
|328452.15|283974.42467985756|
+-----+-----+
only showing top 10 rows
```

Model Evaluation

All three models were evaluated using:

- **RMSE** – Average prediction error in dollars (lower is better).
- **MAPE** – Average error in percentage (e.g., 10% off from actual).
- **Accuracy (%)** – How correct the model is= 100 – MAPE
- **R² Score** – How well predictions fit actual data (closer to 1 = better).

```
=== Model Comparison Summary ===
```

Model	RMSE	MAPE (%)	Accuracy (%)	R ² Score
Linear Regression	68345.60	16.26	83.74	0.1764
Random Forest	66468.18	15.94	84.06	0.2211
GBT Optimized	44260.10	10.01	89.99	0.6581

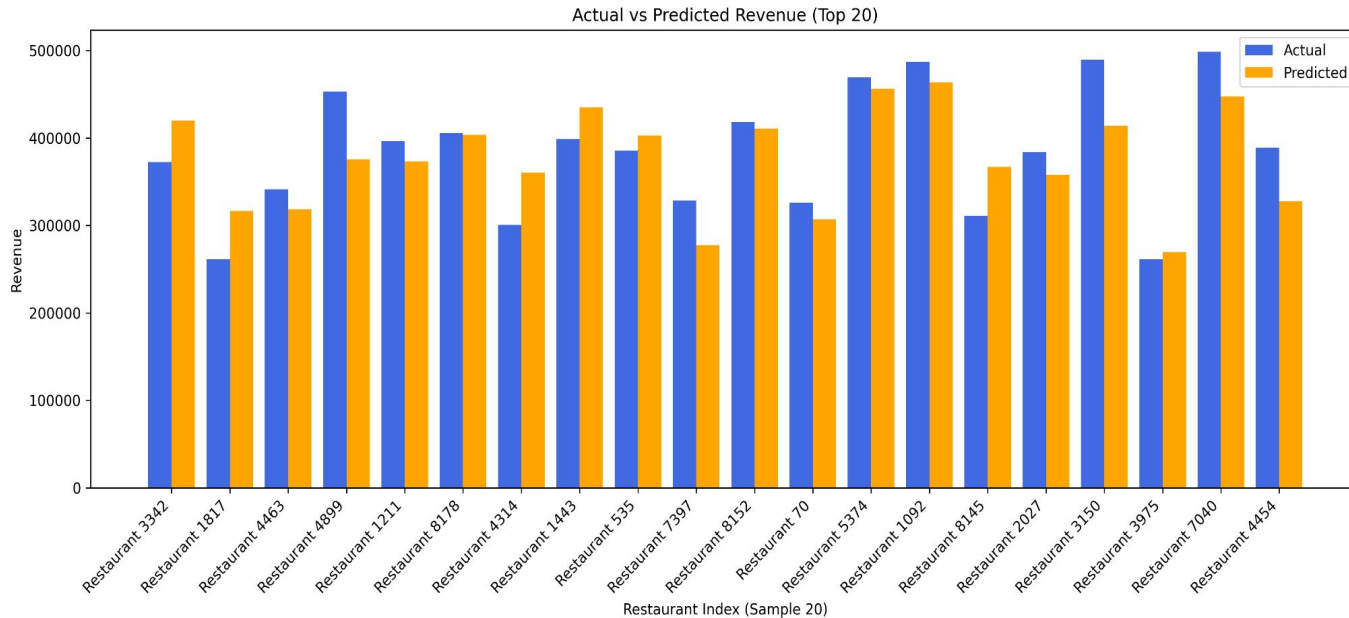
Exported Results from GBT Model

- Full pipeline built using **Apache Spark** and **MongoDB**
- Trained & compared: Linear Regression, Random Forest, GBT
- **Best model (GBT)** reached ~90% accuracy
- Saved to MongoDB collection.

	A	B	C
1	Name	Revenue	prediction_actual
2	Restaurant 3342	372361.37	419907.8197481
3	Restaurant 1817	261326.89	316547.6746266
4	Restaurant 4463	341552.37	318326.0522141
5	Restaurant 4899	452932.93	375482.5884373
6	Restaurant 1211	396465.45	373362.0612598
7	Restaurant 8178	405786.88	403659.0810919
8	Restaurant 4314	300684.68	360166.9939471
9	Restaurant 1443	398573.67	435119.4442224
10	Restaurant 535	385802.15	402908.1596581
11	Restaurant 7397	328452.15	277507.925249
12	Restaurant 8152	418488.55	410527.5511737
13	Restaurant 70	325942.35	306888.4639923
14	Restaurant 5374	469609.27	456218.4097569
15	Restaurant 1092	487246.29	463864.1361662
16	Restaurant 8145	311352.48	366923.9940559
17	Restaurant 2027	384091.92	357678.1003093
18	Restaurant 3150	489317.61	413786.7502091
19	Restaurant 3975	261216.85	269265.712281
20	Restaurant 7040	498544.82	447326.3159445

Bar Chart – Actual vs Predicted Revenue (20 Restaurants)

- Compares **actual vs predicted** revenue side by side for 20 restaurants.
- **Blue bars:** Actual revenue
- **Orange bars:** Predicted revenue



Scatter Plot – Actual vs Predicted Revenue (20 Restaurants)

- Shows point-to-point comparison of actual and predicted revenue.
- **Blue dots: Actual values**
- **Orange dots: Predicted values**
- **Useful to see how close predictions are.**

